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Metamaterials mapped lightweight structures by principal stress lines and topology optimization: Methodology, additive manufacturing, ductile failure and tests

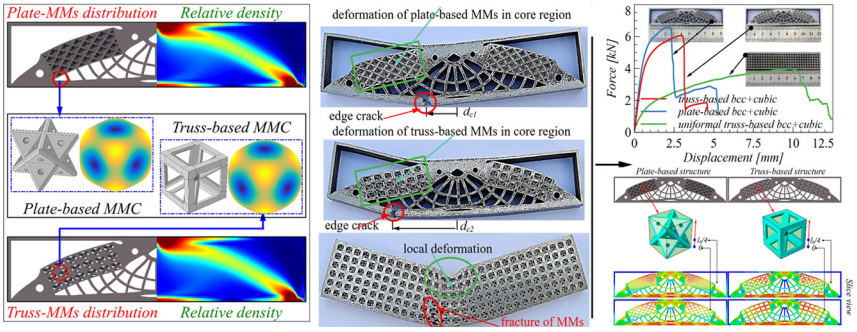
Qingyuan Liu [a,b], Rongwei Xu [b], Yang Zhou [a], Jinguo Ge [b], Shuai Yuan [b,c], Yuhong Long [b,*], Tielin Shi [a,*]

[a] School of Mechanical Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China
[b] School of Mechanical and Electrical Engineering, Guangxi Key Laboratory of Manufacturing Systems and Advanced Manufacturing Technology, Guilin University of Electronic Technology, Guilin, Guangxi 541004, China
[c] School of Materials Science and Engineering, Harbin Institute of Technology, Shenzhen 518055, China

HIGHLIGHTS

- A method of metamaterials mapped multiscale lightweight structures is presented.
- Metamaterials, ribbons, and solid materials are arranged according to PSLs and stress.
- Plate- and truss-based metamaterials mapped lightweight structures are fabricated by SLM.
- Ductile fracture process is investigated numerically and experimentally.

GRAPHICAL ABSTRACT



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ABSTRACT

In lightweight engineering, mechanical metamaterials (MMs) are suitable for additive manufacturing (AM) due to their geometry complexity. However, it is inevitable to bring about mechanical deviation and local yield for MMs infilled lightweight structures. In the current work, MMs mapped structures, named as MMs-TO-PSLs, are proposed. The design domain is divided into core-region, intermediate-region, and solid-region. Core-regions derived from topology optimization (TO) are mapped with MMs, of which the [100] orientations are aligned with the normal directions of principal stress lines (PSLs). The intermediate-regions are infilled with ribbons according to PSLs. The solid regions are allocated to stress concentrated areas. Finally, near isotropic mechanical response is achieved in core-region from anisotropic MMs, and the serious local deformation is eased. To validate such approach, the mapping strategy is carried out on a beam subjected to three-point bending. Metallic MMs-TO-PSLs and MMs infilled uniform lightweight structure are fabricated by selective laser melting (SLM). Elastoplastic simulations with ductile failure model are proposed to investigate the strengthen mechanism of MMs-TO-PSLs. Experimental and numerical results confirmed the ultimate strength of MMs-TO-PSLs can be 40% larger than conventional MMs infilled uniform structures.

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* Corresponding authors.
E-mail addresses: longyuhong@guet.edu.cn (Y. Long), tlshi@hust.edu.cn (T. Shi).

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1. Introduction

The artificial MMs exhibit extraordinary physical features and tunable mechanical properties that cannot be obtained from conventional materials [1–3], and the porous features of MMs bring about weight reduction and high surface area [4,5]. The potential usage of MMs covers lightweight engineering [6–9], biomechanical engineering [10–13], aerospace engineering [14,15], and super heat transfer devices [16]. In mechanical engineering and lightweight fields, MMs have the potential to bring about innovations due to their high mass-specific strength, shock and damage resistance, and damping enhancement properties [17–21]. In the current work, we mainly proposed a MMs-TO-PSLs mapping strategy, by which multiscale lightweight structures with improved strength can be realized by MMs with little mechanical function deviation.

Periodic metamaterial cellular (MMC) is always composed by interconnected basic elements, which include trusses, tubes, plates, shells or their combinations. From the view point of composing elements, MMs can be divided into truss-based [21–23], plate-based [24–26], shells-based [7,27–29] MMs and so on. 2D metamaterials are numerically design by Giorgio and Isola [30,31], elastic surfaces composed by 2D MMs exhibit excellent stretching, bending, and torsion properties. Theoretical simulations suggested that MMs are of high mass-specific stiffness and programmable mechanical properties [24,32]. And plenty of experimental studies [19,21,33–43] have investigated the mechanical features, including elastic properties, mechanical-anisotropy, failure response, plastic behavior, and bulking to yield transformation phenomenon of MMs. The mechanical properties of MMs have a strong relationship with their material volume fraction, and plate-based MMs are thought of high mass-specific strength compared with truss-based counterparts [24,32,44]. On the other hand, with the rapid development of MMs, conventional manufacturing methods encounter problems in the formation of complex MMs. Though complex multiscale lightweight structures can be fabricated by SLM method, the plate-based MMs, belonging to closed-type MMs, cannot be directly fabricated by SLM equipment before the problematic of powder removal is solved [7,27]. In order to tackle such a problem, powder removal hole (PRH) is introduced into plate-based MMs [32]. However, in SLM manufacturing it is time consuming to find a suitable size of PRH. To identify the appropriate size of PRH, a stainless steel 316L sample with gradient values of PRH radius and wall thickness is fabricated by SLM equipment in the current work. Then, both truss-based and plate-based MMs are used for MMs-TO-PSLs mapping process.

Usually, lightweight design can be realized by TO simulations [45]. The introducing of MMs into TO can be taken as a multiscale problem, due to the size of interconnected elements within MMC can be less than 1/100 of the topology optimized geometry. Up to present, lots of researches are proposed on the optimization of lattice-based MMs infilled structures. From the researches of Bendsøe [46,47] the solid isotropic microstructure with penalty model (SIMP) is proposed, and different material interpolation schemes are discussed by applying TO simulation to a 2D lattice-like structure. Rosen proposed a method [48] for lattice-based MMs infilled structures that the MMs are arranged according to the shape of parts. Chang designed an approach of lattice-based MMs infilled structure by using size optimization of the radius of each strut within MMC [49]. From the research of Panesar, et al. [50], a strategy for graded lattice structures controlled by information of TO simulations. Information derived from TO simulations is linked to the variation of MMs. However, TO simulations are usually conducted under isotropic material model, and then anisotropic MMs are filled into design domain basing on TO simulated results. It is inevitable to bring about mechanical deviations between MMs mapped lightweight structures and TO results. On the other hand, by the manipulation of MMs into design domain the local mechanical properties of the MMs mapped structures are rather different from conventional solid materials due to mechanical-anisotropy of MMs. In practical, under the applied load the flaws initiate at the weak directions due to mechanical-anisotropy. In order to tackle such a problem, some researchers conducted approaches in which the struts are arranged according to load directions so that material efficiency can be improved [51,52]. An algorithm based on PSLs and TO simulation for initial structure generation is proposed by Kwok [53], the shape of optimized structure can be refined by density of PSLs. In the researches of Sales [54], toolpath is generated according to PSLs for additive manufacturing and the structure performance is preserved by adjusting the PSLs density. With considering the mechanical-anisotropy of lattices, Stankovic conducted a study on the optimization of lattice structures [55]. Jin [56] carried out investigations on lattice direction by distortion of a coordination system through TO. Wu et al. [57] proposed a novel optimization design method of bone-like porous structure, this kind of lightweight structure reflects stress flow features. In order to improve the mechanical properties of lightweight structures, Bai et al. [58] propose an effective design approach for graded truss-based MMs infilled structures. Under the help of TO simulations, Cheng et al. [59] investigated the variable-density cellular structures for additive manufacturing. It is the effective mechanical properties and the distributions of mesoscale MMs that determine the macroscale mechanical properties of multiscale lightweight structure. It is urgent to design appropriate mapping strategy that cannot only realizes MMs mapping but also shields the mechanical-anisotropy of MMs. Thus, the machinability and effective mechanical properties of mesoscale MMC, TO simulations, and orientation of mapped MMs should be comprehensively considered so that the designed high mass-specific multiscale lightweight structure can be successfully realized by practical SLM manufacturing system.

In the present work, in order to tackle above problems, a MMs-TO-PSLs mapping strategy is presented and MMs are mapped to core-regions derived from TO simulations according to directions of PSLs with considering the machinability of MMs. The main part of the paper is organized as following. In Section 2, the proposed methodology for MMs-TO-PSLs mapping strategy is introduced. In Section 3, the stiffness of plate- and truss-based MMs we used is investigated and a mesoscale mechanics model is constructed for following TO simulations. Besides, the machinability of plate-based MMs is investigated experimentally. In Section 4, with considering the mesoscale mechanical properties of MMs, conventional TO simulations are modified to generate high relative density regions. In Section 5, the MMs mapping strategy is introduced, and the core-region in TO simulations is mapped by MMs with [1 0 0] orientations aligned with the normal directions of PSLs. The stress concentrated regions are infilled by solid materials in case of serious local deformation. In Section 6, elastoplastic FEM simulations conject with ductile fracture evolution are carried out. The values of ultimate strength of MMs-TO-PSLs structures are predicated and fracture mechanism of metallic lightweight structures are numerically investigated. Finally, in section 7 both MMs-TO-PSLs structures and MMs mapped uniform structures are manufactured by SLM equipment. And three-point bend tests are proposed to study the ultimate strength and failure pattern of different multiscale lightweight structures.

2. Methodology for metamaterial mapping strategy

To realize lightweight additive design and manufacturing, the artificial MMC is used as mapping cellular material in the

core-region from structural TO simulations. It notes that the mechanical characteristics of MMC are always anisotropic due to the porous-like distribution of internal elements. The strongest orientation of a MMC can be much stiffer than its weakest direction, structures with controlled anisotropy can be found in [24,27,32,60]. In our study, the [1 0 0] orientations are aligned with the normal directions of PSLs, so that the same orientation of MMs is under heavy load effect. In order to enhance lightweight features, only high-density region from TO simulations will be defined as core-region. To improve material efficiency, the MMs, TO simulations, and PSLs are comprehensively considered. After the design procedure, the MMs mapped lightweight structures are manufactured by SLM system. The mechanism of strength enhancement of MMs-TO-PSLs structures is investigated by elastoplastic FEM simulations and three-point bend experimental tests. The main structure of our work is divided into four parts as shown in Fig. 1.

(a) *MMC preparation*: The investigations MMC, including MMC type, relationship between material volume fraction and stiffness ($E-f_v$ relationship), mechanical-anisotropy, and SLM machinability, are the foundations of both TO and the strengthen of MMs-TO-PSLs structures.
(b) *Topology optimization basing on mesoscale material model*: This relates to the optimization of relative density distribution, which determines the distribution of mesoscale MMs. To reasonably recover the mechanical properties of mesoscale MMs, $E-f_v$ relationship derived in step (a) is used in TO process.
(c) *MMs-TO-PSLs multiscale lightweight structure design*: After TO in step (b), the distribution of intermediate density and stresses can be obtained. In high intermediate density region with uniform stress, MMs are mapped according to PSLs so that the mechanical-anisotropy MMs exhibit near isotropic mechanical response. In high intermediate density regions with heavy stress, solid materials are infilled in case of serious local deformation. Finally, the low intermediate density regions are mapped with ribbons according to PSLs.
(d) *Lightweight structures assessment, SLM fabrication, and mechanical tests*: The ductile failure of lightweight structures under three-point bend effect are investigated by FEM simulations and experimental tests. Small modifications will be introduced during this process before final production.

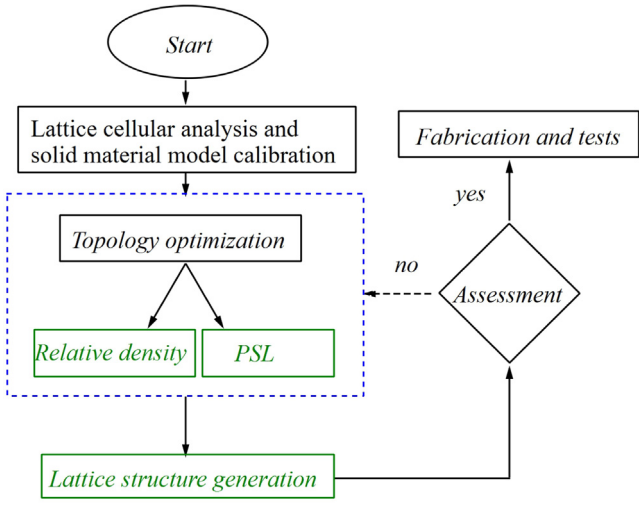


Fig. 1. Schematic of the flowchart of the TO and PSL controlled MMs mapping strategy and AM.

The key part of MMs-TO-PSLs design method is the MMs mapping strategy, of which the major purpose is to identify the distributions of intermediate density, stress and PSLs. Then MMs can be mapped into high-density regions according to PSLs. In the current work, all these capabilities are incorporated into conventional TO simulation software. The simplified introduction of MMs mapping strategy is shown in Fig. 2. As mentioned above, the [1 0 0] orientations of MMs are mainly under load bearing effect. Thus, both the f_v dependent mechanical properties and mechanical-anisotropy of MMs are investigated firstly (Fig. 2(a)). In TO process, to meet computational efficiency as well as to be widely applicable, commercial software is applied to find the high-density regions during TO process. In order to approach the mesoscale mechanics of MMs during TO simulations, the f_v dependent average Young's modulus is used in TO simulations (Fig. 2(b)-(c)). According to distribution of principal stress, the PSLs are traced. Intermediate surfaces are formed by extrusion of the modified PSLs. Subdomains are constructed between neighbor intermediate surfaces (Fig. 2(d)). Then, subdomains are filled by MMs, which are aligned with the normal direction of PSLs. The final MMs mapped lightweight structure is shown in Fig. 2(e).

3. Stiffness of MMC and machinability design

3.1. Mesoscale mechanical model of MMC

Generally, the MMC can be considered as a collection of interconnected basic elements, which can be trusses, plates, shells or their combinations. The strength of final lightweight structures can be influenced by mesoscale mechanical properties of infilled MMs. In the current work, two types of MMC, including truss- and plate-based types as shown in Fig. 3, are used as mapping cellular materials. The MMC can be taken as two-phase material including solid and air. PRH is introduced into plate-based MMC

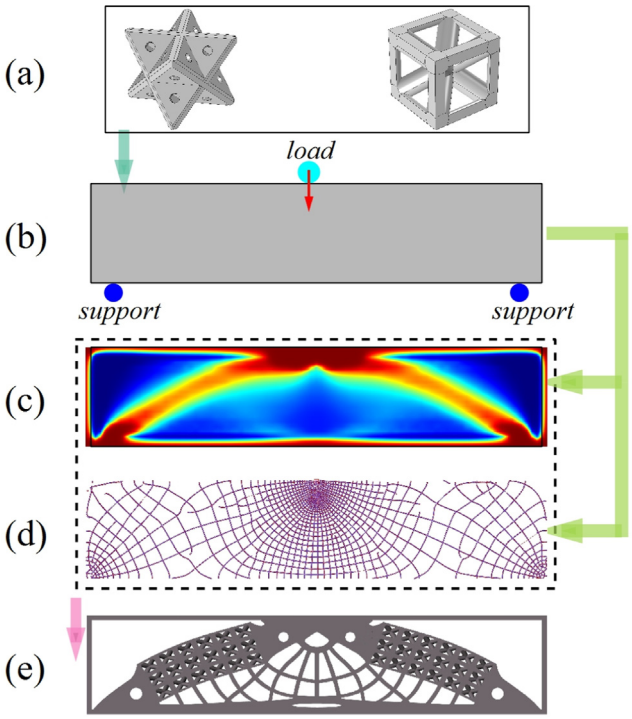


Fig. 2. Illustration of the MMs-TO-PSLs infilling strategy: (a) $E-f_v$ relationship and mechanical-anisotropy of MMs, (b) Mesoscale material model based TO simulation, (c) Information for intermediate density distribution. (d) PSLs distribution in design domain, and (e) MMs infilling basing on intermediate density and PSLs.

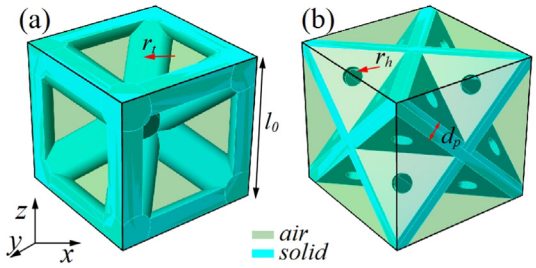


Fig. 3. MMs cellular used in lightweight structures mapping process: a) Truss-based MMC, and b) Plate-based MMC. The solid phase of stainless steel 316L is drawn in blue color, and the air phase is colored in light green. Powder removal holes are introduced into plate-based MMC in case of the problematic of powder removal in SLM process.

in case of the problematic of powder removal in SLM process. Here, the mesoscale mechanics of MMC is expressed as function of f_v, and in following TO simulations the $E-f_v$ relationships are seamlessly used as material models.

The mechanical properties of MMC are only determined by cellular geometry, constituent material type and material volume fraction. For a given material type (stainless steel 316L in our case), the mechanical properties of a MMC are only determined by f_v. This relationship usually is described by the Gibson–Ashby model [61] as shown in Eq. (1).

$$\begin{cases} E = E_0 f_v^{\alpha} \\ f_v = V_{meta}/V_0 \end{cases} \tag{1}$$

where E, E_0, f_v and α are respectively material volume dependent Young's modulus of MMs, Young's modulus of constituting material (200 *GPa*), material volume fraction, and scaling coefficient (larger than or equal to 1.0). The f_v is defined as the ratio of V_{meta} (the volume of solid phase within MMC) and V_0 (total volume of solid and void within MMC). From Eq. (1), it can be found the values of stiffness of MMs are less than their solid constituting materials. It has to be noted the MMs possess the advantages of being lightweight and high specific strength, which cannot be obtained from their constituting solid materials.

Mechanical parameters, including Young's modulus (E), shear modulus (G), bulk modulus (B) and elastically-anisotropy, of MMC can be reflected by elastic constant matrix (C). Both truss- and plate-based MMs we considered are of cubic symmetry. Therefore, the elastic constant matrix can be expressed as [62]:

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44} \end{bmatrix} \tag{2}$$

In Eq. (2), only three independent entries (C_{11},C_{12},C_{44}) should be determined. Theoretically, the value of E is related to flexible matrix (S):

$$S = C^{-1} \tag{3}$$

For MMC of cubic symmetry, the E is orientation-dependent, which can be described in a simple form [63]:

$$E_{orientation} = \frac{1}{(S_{11} + (1-A)S_{44}\left((l_1 l_2)^2 + (l_2 l_3)^2 + (l_3 l_1)^2\right))} \tag{4}$$

where S_{11} and S_{44} are entries in flexible matrix S.

In spherical coordinates, l_1, l_2, l_3 in Eq. (4) can be respectively written as:

$$\begin{cases} l_1 = sin(\theta)cos(\varphi) \\ l_2 = sin(\theta)sin(\varphi) \\ l_3 = cos(\theta) \end{cases} \tag{5}$$

where $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$.

Under the help of linear elastic FEM simulations, the three independent entries, including C_{11},C_{12}, andC_{44} of MMC, can be derived. Then, the stiffness of MMC can be investigated after the C is obtained.

3.2. Micromechanics simulation setup

The entries of C can be derived from the effective stress–strain relationships:

$$\sigma_{eff} = C * \varepsilon_{eff} \tag{6}$$

The effective stress (σ_{eff}) and effective strain (ε_{eff}) values in Eq. (6) are derived from the results of linear elastic FEM calculations. As shown in Fig. 4, the cubic wireframe represents the smallest domain that enclosed the MMC.

Six groups of prescribed loads and appropriate boundary conditions, listed in Table 1, are separately applied to *left*-, *right*-, *front*-, *back*, *top*- and *bottom*-plane. The notation of U_i ($i \in (x, y, z)$) in Table 1 represents the displacement in i-direction. The U_i produces tensile/shear effect when it parallels/perpendiculars to the normal direction of the load applied plane. The *SYMM* in Table 1 means symmetry constraints of the plane.

3.3. $E-f_v$ relationship and mechanical-anisotropy

The $E-f_v$ relationships are obtained by changing the values of f_v, which is controlled by regulating truss radius or plate thickness. Under different values of f_v, the independent elastic constants of truss- and plate-based MMs are presented in Fig. 5(a)-(b). The values of C_{11}, C_{12}, and C_{44} are respectively presented in red, blue, and green bars. From a rough observation, it can be found the entries of C for plate-based MMs are greater than their truss-based counterparts under similar value of f_v.

Mechanical-anisotropy of MMC can be learned by both $E_{orientation}$ (Eq. (4)) and Zener's anisotropy ratio [64]. For MMC with cubic symmetry, the Zener's anisotropy ratio can be written as:

$$A = 2(S_{11} - S_{12})/S_{44} \tag{7}$$

The unify of A relates to the elastic-isotropy of a material. For truss- and plate-based MMC, the Zener's anisotropy ratio and $E_{orientation}$ are shown in Fig. 6(a)-(b). The extent of mechanical-anisotropy of these MMs decreased with the increase of f_v. From Zener's anisotropy ratio, it also can be found mechanical-anisotropy of truss-based MMs becomes less obvious when f_v is greater than 60%. For plate-based MMs, the mechanical-anisotropy also is not obvious when f_v is greater than 60%. From

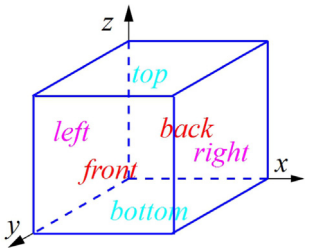


Fig. 4. Illustration of boundaries of the cubic domain that enclosed MMC.

Table 1
Boundary conditions for the calculation of elastic constant matrix of MMC.

BCs	Right	Left	Front	Back	Top	Bottom
1	$U_x = \varepsilon_0$	SYMM	$U_y = 0$	SYMM	$U_z = 0$	SYMM
2	$U_x = 0$	SYMM	$U_y = \varepsilon_0$	SYMM	$U_z = 0$	SYMM
3	$U_x = 0$	SYMM	$U_y = 0$	SYMM	$U_z = \varepsilon_0$	SYMM
4	$U_y = U_z = 0$	$U_y = U_z = 0$	U_x=ε_0,$U_z = 0$	$U_x = U_z = 0$	$U_z = 0$	$U_z = 0$
5	$U_y = Uz = 0$	$U_y = Uz = 0$	$Uy = 0$	$Uy = 0$	U_x=ε_0,$U_y = 0$	$U_x = U_y = 0$
6	$U_x = 0$	$Ux = 0$	$Ux = Uz = 0$	$Ux = Uy = 0$	$U_x = 0$,U_y=ε_0	$U_x = U_y = 0$

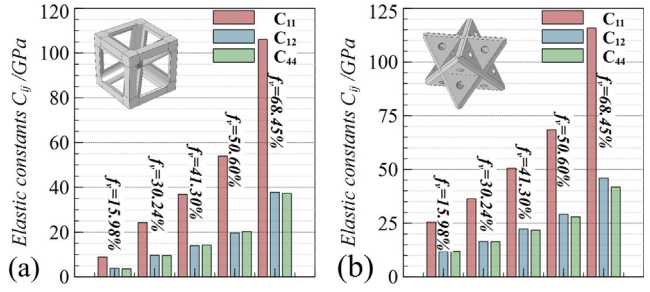


Fig. 5. Independent elastic constants including C_{11}, C_{12}, and C_{44} of a) truss-based MMC, and b) plate-based MMC under different values of f_v.

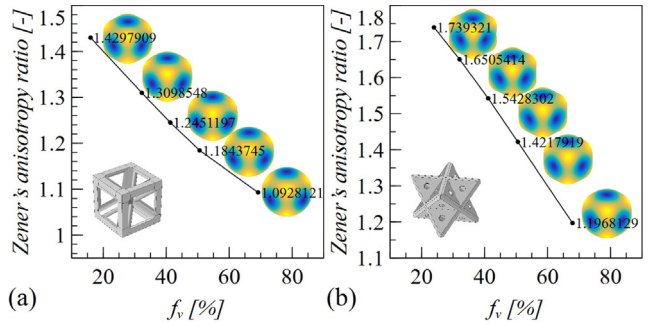


Fig. 6. Mechanical-anisotropy of MMs under different values of f_v. a) Orientation dependent Young's modulus and Zener's anisotropy ratio of truss-based MMC, and b) Orientation dependent Young's modulus and Zener's anisotropy ratio of plate-based MMC.

our investigation in following sections, it hints that the slight mechanical-anisotropy can be ignored by regulating the orientations of MMC. And near isotropic mechanical response can be obtained so that mechanical deviation between MMC and sloid material can be neglected.

Due to the mechanical-anisotropy of MMs, the average Young's modulus ($E_{average}$) will be used to represent the stiffness of MMs. As shown in Fig. 7, the values of $E_{average}$ and Young's modulus in [100] orientations (E_{100}) are plotted as a function of f_v. It can be seen the value of E_{100} is larger than $E_{average}$ under the same f_v. Compared with truss-based MMs, the plate-based MMs have relatively high values of E_{100}. However, it has to be noted that after SLM fabrication the practical strength of MMs may be weaker than theoretical values due to surface and internal flaws.

The $E_{average}$ of MMs is taken as material models in TO simulations, which can be described by f_v as:

$$\begin{cases} E^{truss}_{average} = 41.07 * \exp(1.758 f_v) - 38.98 \\ E^{plate}_{average} = 21.15 * \exp(2.355 f_v) - 21.0 \end{cases} \quad (8)$$

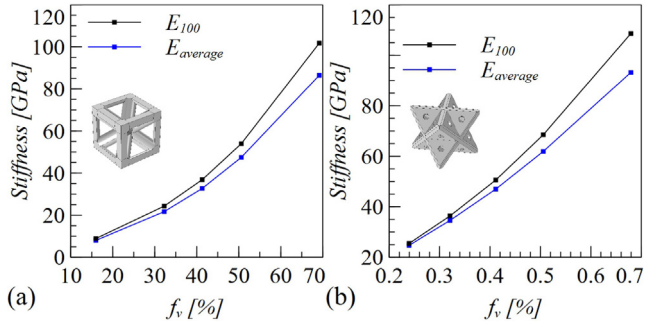


Fig. 7. Average Young's modulus and Young's modulus in [100] orientation of MMs. a) Truss-based MMs, and b) plate-based MMs under different values of material volume fraction.

It notes the f_v of mapped MMs is about 60% so that the mechanical-anisotropy of MMs becomes less obvious. After TO simulations, the MMs are mapped to core-region according to PSLs so that the [100] orientations of MMs are mainly under load effect. Thus, the material efficiency of MMs can be improved due to shear effect is eased in this mapping strategy.

3.4. Machinability design and experimental tests of MMs

The machineability should be considered before practical SLM fabrications. First of all, the smallest characteristic size of internal elements within MMC should be larger than machining resolution of SLM system. Then, the designed MMs should meet the requirement of SLM. The machineability of conventional truss-based MMC is not discussed here, due to its open-type and self-supporting geometrical features. We mainly investigate the machineability of plate-based MMs, due to they belong to closed-type MMs. In order to tackle the powder removal problem, PRHs are introduced into plate-based MMs in design procedure (in Fig. 3(b)).

The machineability of plate-based MMs with PRHs is directly examined by practical SLM fabrications. To reduce testing costs, as shown in Fig. 8 the SLM fabricated plate-based samples are composed by graded MMs. In these samples, the radius of PRHs ranges from 200 to 500 μm, and the plate thickness ranges from 240 to 320 μm. These graded samples are successfully manufactured by SLM system.

As shown in Fig. 9, additional experiments are carried out to study the effect of PRHs on the strength of plate-based MMs. Under the same plate thickness, it can be found the strength of MMs decreased with the increase of the radius of PRHs. In order to preserve sufficient structural strength as well as to fulfill the requirement of powder-removal, the radius of PRH is set to 300 μm in following SLM fabrications. Under same conditions, the MMs mapped multiscale lightweight structures are fabricated by SLM system.

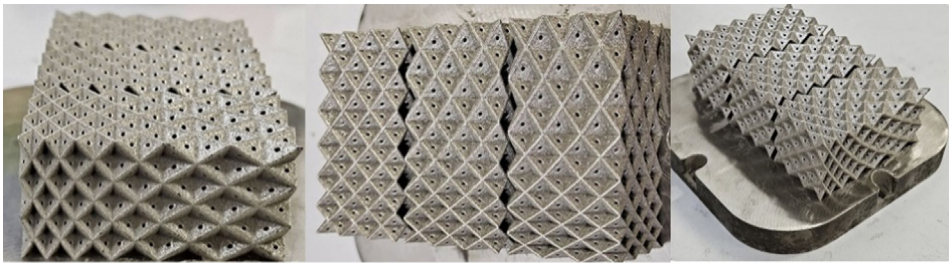


Fig. 8. Machinability design and SLM fabrication.

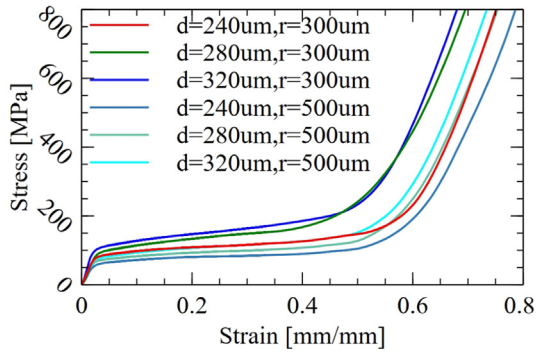


Fig. 9. Engineering stress–strain response for periodic MMs samples under compression load.

4. MMs mapping strategy basing on topology optimization and PSLs

TO simulations are used to produce relative density information, while PSLs are applied to construct subdomains in high-density region. Then, MMs are mapped to subdomains, and [100] directions of MMs are automatically aligned with the normal direction of PSLs. This section contains three subsections: 1) TO simulations with modified material model. 2) Trace of PSLs and construction of subdomains. 3) MMs infilling.

4.1. Structural topology optimization basing on different material models

Belonging to the optimization problem, the TO method, containing objective function and design constraints, is usually solved in a discrete FEM design domain. Strain energy based TO is adopted, and the basic expressions of TO method is given as:

$$\begin{cases} \text{Minimize} c(\rho) = \dfrac{U^T K U}{2} \end{cases} \tag{9}$$

$$V_{target} = \sum_{i=1}^{n} V_i \rho_i \tag{10}$$

In Eq. (9), $c(\rho)$, K, and U are respectively the objective function, stiffness matrix, and displacement vector. In Eq. (10), V_{target}, V_i, and ρ_i are respectively the total material volume of all FEM elements, the volume of FEM element i, and the density of FEM element i. It notes that ρ_i is a continuous number between 0.001 and 1. The relationship between K and U is described by $KU = F$ during on solid mechanics FEM calculations. F is force. During the numerical iteration process of TO, each FEM element can be taken as a new material, due to the mesoscale mechanical properties changed with ρ_i. TO simulations are proposed basing on solid isotropic material penalization (SIMP) method [65].

In SIMP based TO, the intermediate stiffness matrix (C^i) of local material is expressed as:

$$C^i = C_0 * f(\rho_i) \tag{11}$$

In Eq. (11), $f(\rho_i)$ represents the shrinkage of stiffness (C_0) of constituting materials, which can be described as:

$$f(\rho_i) = f_v^{\varepsilon} \tag{12}$$

In expression (12), ε is the scaling coefficient, which is usually larger than 1.0. The SIMP provides an artificial material model that describes the f_v dependent mechanical properties. As a numerical approach, the SIMP model does not have a strong physical meaning of f_v dependent mechanical properties. Besides, the value of ε should be carefully tested before final TO simulation.

In Eq. (11), C^i is directly described by Eq. (8), which is relative to f_v. Under the constraint of target volume ($V_{target} = 0.35$ in our case) and objective function (Eq. (9)), the optimal intermediate density distribution can be obtained.

4.2. Relative density distributions under different material models

For a comparison purpose, both the SIMP model (Eq. (11) and Eq. (12)) and material models (Eq. (8)) derived from mesoscale mechanical simulations are presented in Fig. 10. With the increase of ε in Eq. (12), it can be expected that the curves tend to be nonlinear and approach to the profile of step function. This indicates that intermediate materials turn out to become void or solid with the increase of ε. However, it takes time to identify an appropriate ε in Eq. (12). While the Eq. (8) directly provides a link between mesoscale mechanical properties and material model in TO simulations.

Under different material models, after TO simulations the relative density distributions and von Mises stresses are shown in Fig. 11. The results in Fig. 11(a)-(b), Fig. 11(c)-(d), and Fig. 11(e)-

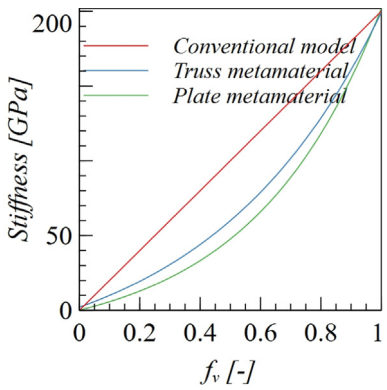


Fig. 10. Comparison of $E - f_v$ relationships between conventional and mesoscale material models used in topology optimizations.

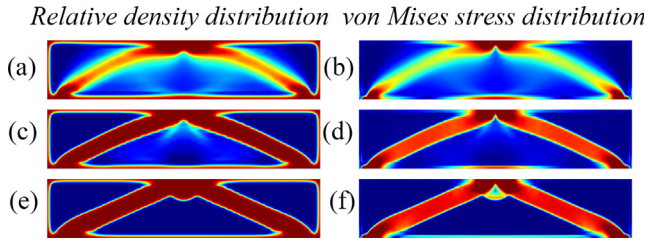


Fig. 11. Relative density and von Mises stress distribution under different material models.

(f) are respectively corresponding to conventional SIMP model, truss-based MMC model, and plate-based MMC model. The penalization factor of all models is set to 1.0, after 100 iterations it can be observed the clear boundary of high-density region for truss- and plate-based MMC models. Though material models derived from mesoscale mechanical properties lead to fast converge for TO simulations, the final distributions of high-density regions derived from different models are of similar shapes. It also can be observed stress concentrated at load applied region and supporting positions. However, the stresses and intermediate densities in vast majority of high-density regions are uniformly distributed.

5. MMs-TO-PSLs mapping strategy

After TO simulations, the design domain is divided into high relative density regions (the red regions in Fig. 12(a)) and void-regions (the blue regions Fig. 12(a)). The high relative density regions around supporting point and load position are named as solid region and will be infilled with solid material in case of heavy stress concentration (Fig. 12(b)). Other high relative density regions are core-region and will be mapped with MMs. Before MMs mapping process, the PSLs are traced (Fig. 12 (*d*)) according to principal stress vectors (Fig. 12(c)) and then modified (Fig. 12 (e)). The MMs mapping is realized by allocating MMs along PSLs (Fig. 12(f)). It can be expected the material efficiency of MMs can be improved if the relatively stiff orientations are aligned with the normal direction of PSLs.

5.1. Construction of principal stress lines

Under three-point bend effect, the stress state of beam can be investigated from in plane stress. The 2D solid continuum region can be divided into a collection of infinite small subregions, and the stress state at the center of each subregion can be described by normal and shear stress components (σ_{xx}, σ_{yy}, and τ_{xy}) as shown in Fig. 13. The normal stress components are always corresponding to planer orientation of the small subregion before and after rotation.

The stress state is numerically derived from force balance of local material. After rotation, coordinate transformation of stresses can be described as:

$$\sigma' = \begin{bmatrix} \sigma'_{xx} & \tau'_{xy} \\ \tau'_{xy} & \sigma'_{yy} \end{bmatrix} = D\sigma D^T \tag{13}$$

$$D = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix} \tag{14}$$

$$\sigma = \begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{bmatrix} \tag{15}$$

where θ and σ' are respectively rotation angle stress state after rotation. The principal stresses can be obtained when the value of τ'_{xy} equals zero.

$$(\sigma_{yy} - \sigma_{xx})sin\theta cos\theta + \tau_{xy}(cos^2\theta - sin^2\theta) = 0 \tag{16}$$

Thus, the principal stresses can be obtained when $\theta = \theta_p$, which can be derived from Eq. (16).

$$\theta_p = 0.5\tan^{-1}(\frac{0.5\tau_{xy}}{\sigma_x - \sigma_y}) \tag{17}$$

From above expressions, it can be noticed that at the principal directions only normal stress components take effect. By substituting Eq. (17) into Eq. (13), the principal stress components (the normal stresses when shear stress equals 0) can be described as:

$$\sigma_{max,min} = \frac{\sigma_x - \sigma_y}{2} \pm \sqrt{\frac{\sigma_x - \sigma_y}{2} + {\tau_{xy}}^2} \tag{18}$$

The PSLs described the stress flow under applied load. PSLs provide lines of desirable material continuity in a design domain. For a given material, the PSLs are only affected by shape of design domain, loading and supporting locations. It can be reasonably expected material efficiency of MMs can be improved when appropriate directions of MMs are aligned with normal directions of PSLs.

Stress components are obtained from FEM simulations. By substituting stress components into equations (17) and (18), both the

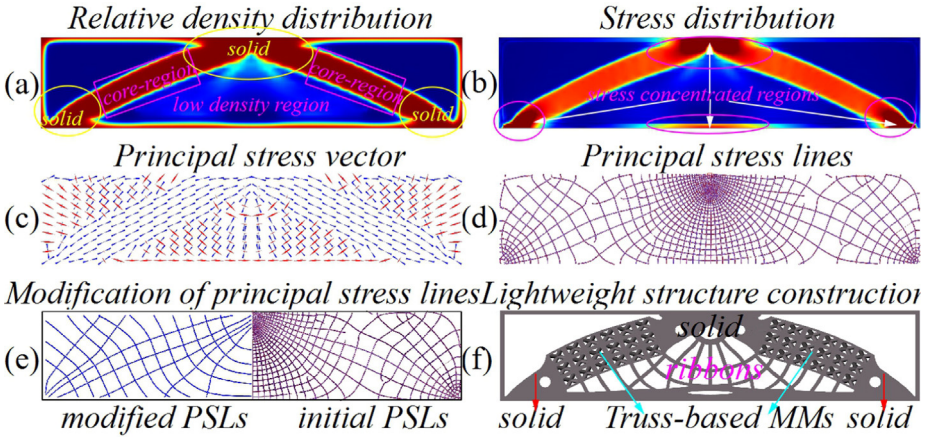


Fig. 12. Topology optimization and MMs infilling according to principal stress lines.

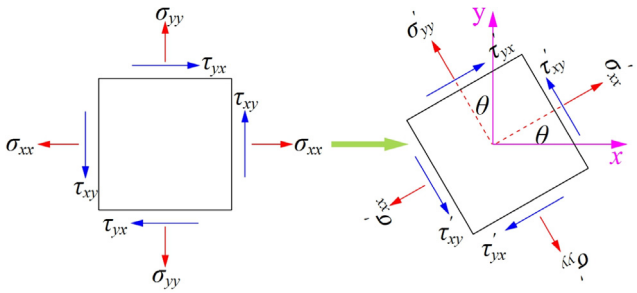


Fig. 13. Illustration of stress components before and after rotation.

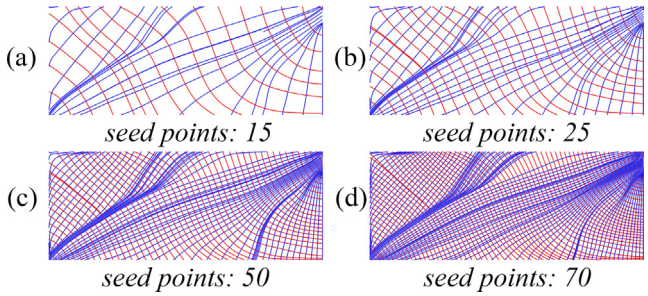


Fig. 15. Variation of the density of PSLs by regulating seed points.

amplitudes and orientations of principal stresses are derived. For the three-point bend beam case, the PSLs vectors are shown in Fig. 14.

PSLs are generated according to the point cloud of PSLs vectors. Usually, a PSL can be traced from a starting point in the principal direction. By moving a small step along current principal direction, we can find the next point incrementally along the PSL. Assuming the length of boundary is l_b and the number of seed points is n, the boundary can be uniformly divided with a l_b/n neighbor distance. From stress distributions, it can be found stress mainly distributed along the line from supporting point to load region. Therefore, uniform seed points are allocated from the supporting point to load point. Due to the symmetry features of design domain, only a half model is presented here. By changing the number of seed points from 15 to 70, the density of PSLs is varied accordingly (Fig. 15(-a)-(d)). In SLM fabrication, the length of MMC cellular we selected is about 3 *mm*. Thus, 10 seed points are dense enough to represent principal stress state of the model.

The initial PSLs are shown in Fig. 16(a). These PSLs are not appropriate for MMs mapping due to their poor shapes and bad continuity. There are mainly three problems should be tackled: 1) Short PSLs should be removed, due to they are not appropriate for MMs mapping. 2) Several PSLs may be around one seed point, only one PSL should be left behind. 3) The PSL may be turning back, and it is not suitable for constructing subdomains. To modify initial PSLs, an approach is provided as shown in Fig. 16. Firstly, all initial PSLs are trimmed by boundary edges, only PSLs within the green region are left behind (Fig. 16(b)). Then, in Fig. 16(c) PSLs are smoothed to avoid sharp corners. In Fig. 16(d), circular searching regions are constructed at the seed points, the radius of circle regions ranges from 0.001 to 0.01. At the same time, intersection points are generated between leading line and PSLs. Only one intersection point will be left behind within each searching region. It notes each intersection point refers to a unique PSL. During this process, PSL will be removed if its length is less than $1/m$ (m greater than 6) of leading line. The reconstructed PSLs are shown in Fig. 16(e).

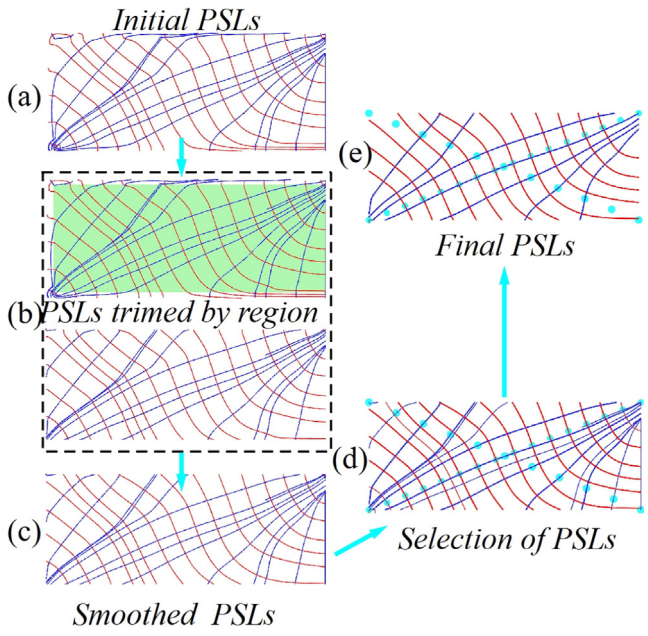


Fig. 16. Reconstructing of principal stress lines.

5.2. MMs infilling in core-region

Under the help of TO and PSLs, the flowchart of MMs mapping in core-region are shown in Fig. 17. The distribution of f_v is shown in Fig. 17(a). If the core-region is defined as region with $f_v > 0.35$, the core-region can be expressed by the light blue region in Fig. 17 (b). Only PSLs within core-region are left for use as shown in Fig. 17 (c). Along the normal direction of core-region, PSLs in red color are extruded to surfaces (in Fig. 17(d)). In Fig. 17(e), near cubic subdomains are constructed between neighbor surfaces. Then, MMs are conformally mapped into these subdomains as shown in Fig. 17 (f). The [100] orientations are mainly aligned with the normal direction of PSLs.

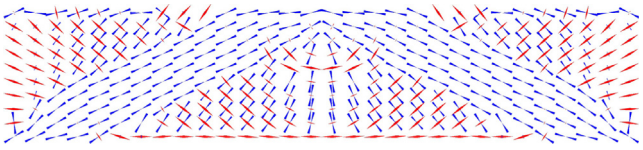


Fig. 14. Principal stress amplitudes and orientations.

5.3. Formation of MMs-TO-PSLs structure

The formation of MMs-TO-PSLs structure will be introduced here. According to simulation in Fig. 11, the stress concentrated at load applied region and supporting positions. Therefore, the high stress regions are mapped with solid materials in case of heavy stress concentration. The stresses and intermediate densities in vast majority of high-density regions are uniformly distributed. MMs will be mapped into high-density regions with relatively low stresses. It notes that the mechanical-anisotropy of MMC is rather weak when f_v is larger than 60% (in Fig. 6). Thus, the mapped MMs are of same f_v, which equals 60%. In this way, the near isotropy mechanical properties can be realized in core-region. Accordingly, initial core-region for MMs-TO-PSLs structures is defined as areas with $f_v > 0.60$. In case of the machining limita-

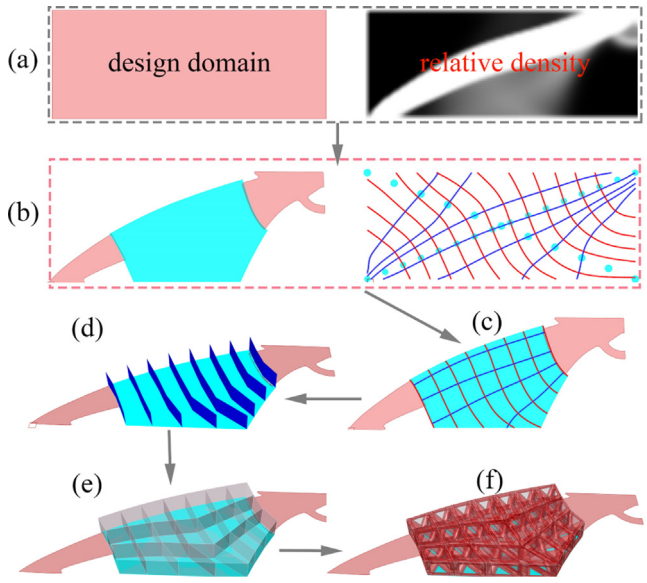


Fig. 17. Flowchart of MMs mapping in core-region under the help of TO and PSLs.

tion of SLM equipment, here the MMs infilled area is set about 1.5 times wide of the width of initial core-region. And the regions with f_v ranging from 0.30 to 0.60 are infilled with ribbons according to PSLs. The final truss-based MMs-TO-PSLs structure is shown in Fig. 18.

6. Elastoplastic simulation of MMs-TO-PSLs structures

Under elastoplastic simulations, different patterns of deformation are compared between MMs-TO-PSLs and MMs mapped uniform structure. Ductile failure model is used to investigate to fracture mechanism of samples under three-point bend effect. The load-deformation response is investigated. Three stages, including elastic, plastic, and fracture propagation, are presented. The mechanism of high stiffness and strength of MMs-TO-PSLs structures is considered.

6.1. Numerical model setup and material properties calibration

To investigate the mechanical behavior of MMs-TO-PSLs structures, numerical simulation model of three-point bend process is carried out. The numerical model is divided in to boundary region (black color) and porous material region (green color). The model sizes are represented by characteristic length ($l_0 = 3mm$) of MMs as shown in Fig. 19(a). The porous material region is $30 \times 6 \times 2$ MMs long in x-, y-, and z-direction. The width of boundary region is set to 1/3 of the characteristic length of MMs cellular. Under the MMs-TO-PSLs mapping approach introduced in sections (2)-(5), plate- and truss-based MMs-TO-PSLs samples are constructed as shown in Fig. 19(c)-(d). For a comparison purpose, under the same volume fraction a truss-based MMs mapped uniform sample (Fig. 19(b)) is also constructed and simulated.

The material properties used in simulations are derived from experimental tests. The engineering strain–stress curve of stainless steel 316L is shown in Fig. 20. The true strain–stress curve is calibrated according to the relationships between true- and engineering-stress (strain).

$$\begin{cases} \sigma_T = \sigma(1+\varepsilon) \\ \varepsilon_T = ln(1+\varepsilon) \end{cases} \tag{19}$$

In Eq. (19), σ, σ_T, ε, ε_T are respectively engineering stress, true stress, engineering strain, and true strain. In elastoplastic simulations, the plastic mechanical properties of stainless steel 316L are described by true stress-stain relationship. Other parameters are the same as elastic simulation in Section 3.1.

The boundary conditions and applied load are applied as Fig. 19 (a). In the current work, the compression effect is produced by applying a constant compression velocity of $10mm/s$. Compression displacement, reaction force and stress distributions are recorded for further analysis.

6.2. Mechanism of structural strengthen and ductile failure of MMs-TO-PSLs

6.2.1. Deformation pattern and stress state

For plate- and truss-based MMs-TO-PSLs structures, the von-Mises stress distributions conject with deformations are respectively shown in Fig. 21(a)-(b). The large stresses are mainly distributed within two regions, including MMs infilled core-region and the solid lower boundary. Compared Fig. 21(a) with Fig. 21 (b), the stresses of plate-based MMs-TO-PSLs structure are more uniformly distributed than that its truss-based counterpart. It contributes to the strengthen of plate-based structures due to the ease of early local yield of MMs. For a comparison purpose, the stress distribution of truss-based MMs mapped uniform structure is shown in Fig. 21(c). Different from MMs-TO-PSLs structures, severe local deformations can be found around the load applied region. The deformation pattern of MMs-TO-PSLs structures is significantly different from truss-based MMs mapped uniform structure. It can be expected the stiffness of truss-based MMs mapped uni-

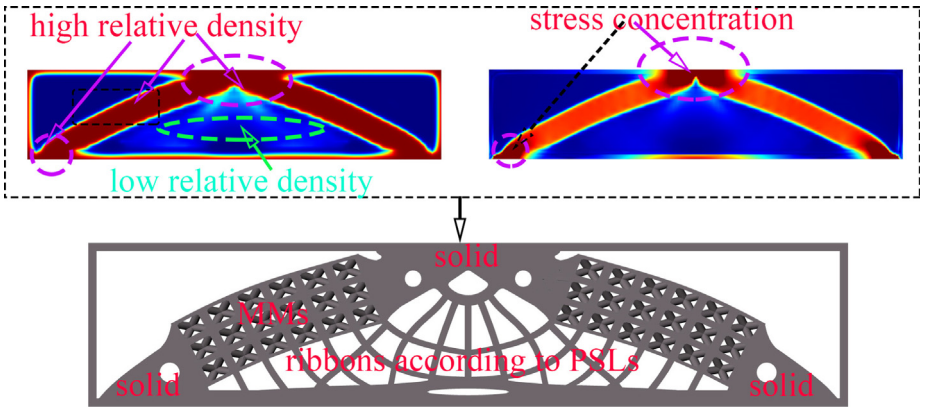


Fig. 18. MMs-TO-PSLs lightweight structures designed under help of TO simulations and PSLs.

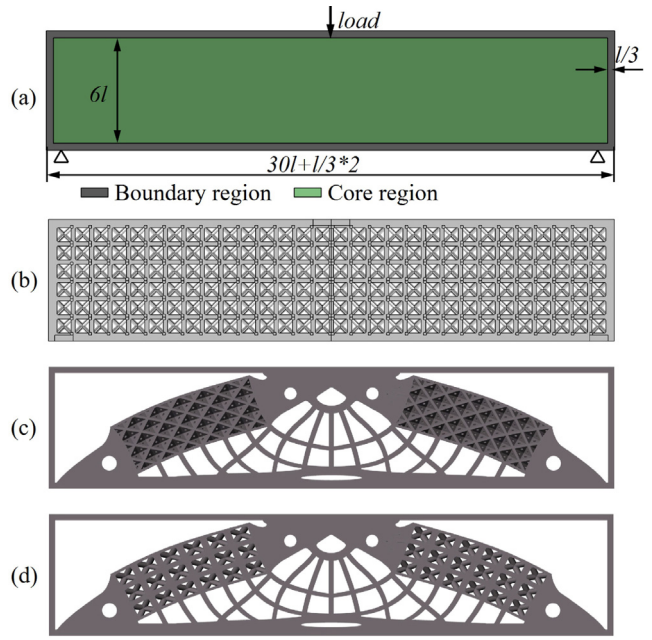


Fig. 19. Final lightweight structures in elastoplastic investigation. a) Illustration of model size and boundary conditions. b) MMs mapped uniform structure. c) Plate-based MMs-TO-PSLs structure. d) Truss-based MMs-TO-PSLs structure.

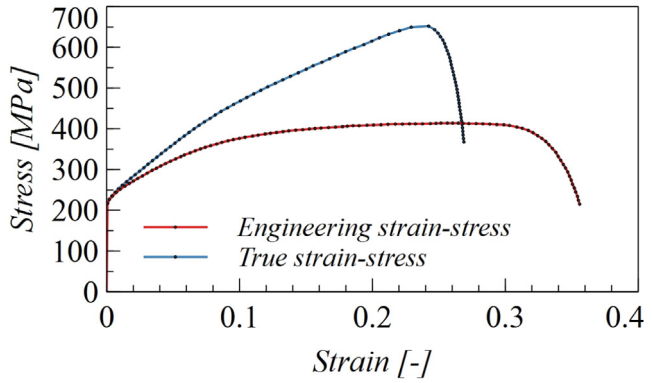


Fig. 20. Calibration of strain–stress relationship used for elastoplastic simulations.

form structure is the smallest among these multiscale lightweight structures, owing to the severe local yield deformation.

For ductile materials, fracture initiates at tensile or shear stresses concentrated regions. Instead of von-mises stress, the distributions of tensile and compression stresses within plate- and truss-based MMs-TO-PSLs structures are shown in Fig. 22(a)-(b). From the distribution of stress state, each MMs-TO-PSLs structure can be mainly divided into two regions, containing compression- and tensile-dominated areas. Moreover, several tensile stress concentrated positions are found and marked by magenta circles. It can be anticipated that rupture is more likely to produce around these positions. In compression-dominated regions, it can be directly observed that the [100] orientations of MMs are mainly under compression effect. Therefore, the shear effect can be eased and the mechanical-anisotropy MMs can exhibit near isotropic mechanical response, due to only one orientation is under compression load effect.

6.2.2. Ductile failure of multiscale lightweight structures

The absolute value of the reaction force (F_R) at the load applied region equals to the external force (F_{ext}), and the overall stiffness (S) of lightweight structures can be evaluated by the reaction force (F_R^{linear}) and compressed displacement (L) at linear elastic stage as:

$$S = \frac{d\left|F_R^{linear}\right|}{dL} \tag{20}$$

From Eq. (20), the stiffness of structures can be reflected by the gradient of linear elastic stage of force–displacement curve. The response of displacement vs. reaction forces for truss- and plate-based MMs-TO-PSLs structures are illustrated in Fig. 23. The results for truss-based MMs mapped uniform structure are also presented for a comparison purpose. The displacement-force curve can be divided into three stages, including elastic, plastic, and rapture stages. From a rough observation, it can be found that truss- and plate-based MMs-TO-PSLs structures are much stiffer than truss-based MMs mapped uniform structure. It also can be observed the plate-based MMs-TO-PSLs structure is obvious stiffer than its truss-based counterpart. This is in agreement with previous elastic properties investigations. In plastic stage, the ultimate reaction force is reached before the rapture happens. It is the yield of local MMs that leads to the decrease of reaction force. For truss- and plate-based MMs-TO-PSLs structures, once the initial MMs are seriously absent from their original positions, stiffness degradation turned out to be obvious. The same reaction force can be produced at low compression displacement for stiff structures. Therefore, it can be found the rapture happens at smaller compressed displacement for plate-based MMs-TO-PSLs structure. After the ultimate force occurs, the amplitude of reaction force suddenly dropped-off due to the fracture of lower solid region. In the rapture stage, the successive fracture of solid ribbons resulted in several drops of force curves before totally fracture occurs.

The strengthen mechanism is studied from the slice view of core-regions. Due to symmetry features of MMs structures in the depth direction, the slice views of von-mises stress for truss- and plate-based MMs-TO-PSLs structures are selected at bottom and $l_0/4$ depth of a MMC as shown in Fig. 24. It can be clearly observed the stresses mainly distributed in core-regions and lower boundary. This is in agreement with the stress distributions in TO simu-

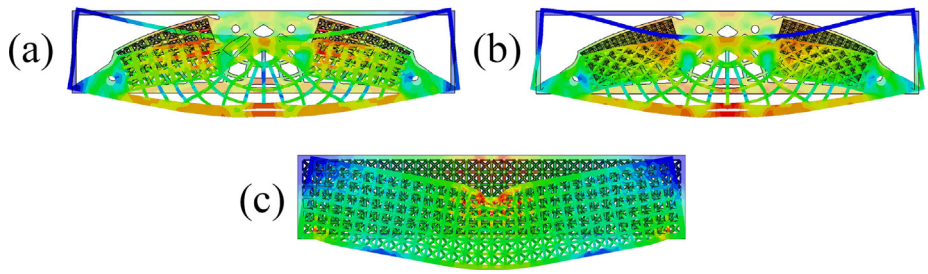


Fig. 21. Von-mises stresses distribution and deformation before and after the three-point bend effect. a) plate-based MMs-TO-PSLs structure, b) truss-based MMs-TO-PSLs structure, c) truss-based MMs infilled uniform structure.

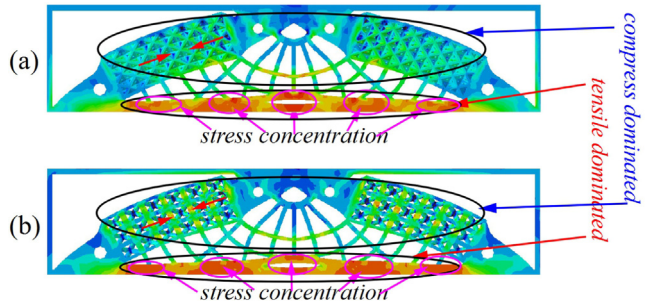


Fig. 22. Compression and tensile stresses distribution of a) plate-based MMs-TO-PSLs multiscale lightweight structure, and b) truss-based MMs-TO-PSLs multiscale lightweight structure under three-point bend effect. The tensile stress concentrated positions are marked by magenta circles.

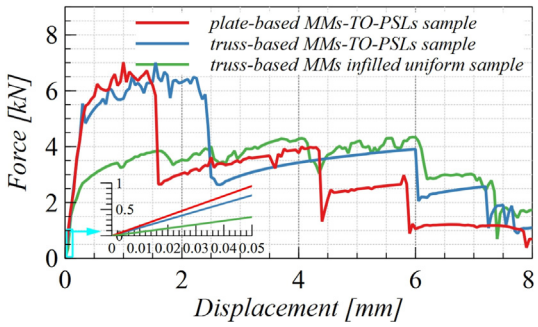


Fig. 23. Numerical simulated force–displacement response for plate (truss)-based MMs-TO-PSLs structure and conventional truss-based MMs infilled structure. Each curve can be dived into three stages, including linear-elastic, plastic, and fracture stages, and the sudden drop-off of force–displacement curve relates to the fracture of materials.

lations. Moreover, it can be found heavy interaction of internal plates within plate-based MMs. It is the strong interaction of composing plates that leads to the improvement of strength of MMs-TO-PSLs structure.

7. SLM fabrication and three-point bend tests

All MMs-TO-PSLs structures are fabricated by a SLM equipment using SS316L powders on metallic substrates with same constituting materials. The radius of metal powders ranging from 30 to 60 *um* (Fig. 25). The chemical content of SS316L powders is shown in Table 2.

The SS316L powders are melted by scanning laser beam in *Ar* gas environment. In SLM manufacturing process, multiscale lightweight structures are printed layer by layer with 0.03 *mm* layer thickness (t_1) and 0.05 *mm* hatch distance (t_2). The printed average laser power (p) is 180*W*. All structures are printed at a scan speed (v) of 1000*mm*/*s*. The details about machining parameters of SLM fabrication system we selected are shown in Table 3.

After SLM processing, metallic samples are removed by wire cutting. The final truss- and plate-based MMs-TO-PSLs samples are respectively shown in Fig. 26(a) and Fig. 26 (*b*), and conven-

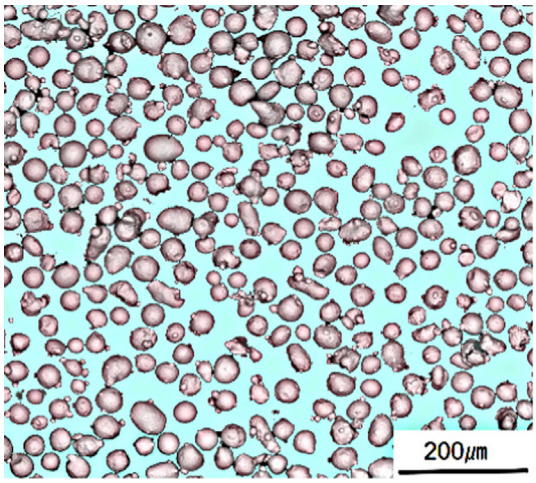


Fig. 25. Distribution of stainless steel 316L powders.

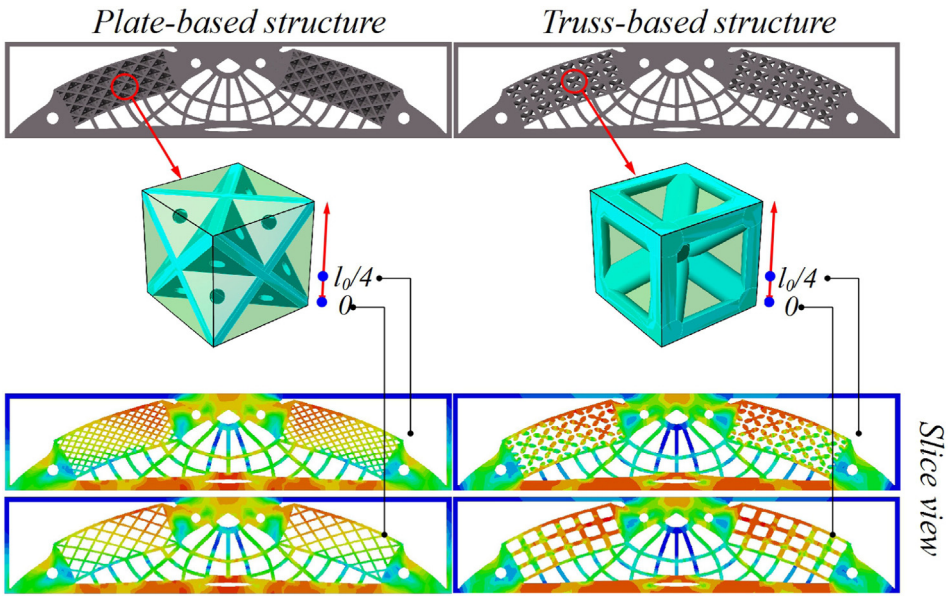


Fig. 24. Slice views of von-mises stresses distribution in a) plate-based MMs-TO-PSLs structure, and b) truss-based MMs-TO-PSLs structure. The slice views are selected at 0 and 1/4 depth of a MMC due to symmetry features.

Table 2
Chemical content of stainless steel 316L [*wt%*].

Si	Cr	Ni	Mn	Mo	C	S	P	Fe
0.64	16.79	11.07	0.68	2.53	0.027	0.0056	0.022	other

Table 3
SLM fabrication parameters.

$p(W)$	$v(mm/s)$	$t_1(mm)$	$t_2(mm)$
180	1000	0.03	0.05

Table 4
Net weight of MMs mapped stainless steel 316L samples (units: *g*).

Truss-based MMs-TO-PSLs	Plate-based MMs-TO-PSLs	Uniform structure
43	47	45

tional truss-based MMs infilled uniform sample is shown in Fig. 26 (c). All samples are of similar net weight as shown in Table 4.

Three-point bend tests (in Fig. 27) are carried out to investigate elastoplastic properties and ductile failure evolution process of these lightweight structures. The compression effect is realized by applying compression displacement from the cylinder punch at the center of upper boundary of lightweight structures. All tests are performed until the fracture of samples. The compression force is measured by a load cell connected to the data acquisition equipment. During compression process, both the displacement of load cell and compression reaction force are recorded for further analysis.

The final fractured structures of plate (truss)-based MMs-TO-PSLs, and MMs mapped uniform samples are respectively shown in Fig. 28(a)-(c). In agreement with numerical simulations, as shown in Fig. 28(c) it can be seen obvious local deformation around load applied region for MMs mapped uniform sample, due to serious compression induced ductile failure of local MMs. At the same time, the tensile stress concentrated regions we previously simulated around the lower boundaries are fractured in three-point bend experimental tests. The symmetry structure is destroyed with the increase of compression displacement, which leads to the edge crack absenting from center of lower edge. For plate-based MMs-TO-PSLs and MMs mapped uniform samples, it can be found the final structure is nearly symmetry, and the edge crack produced around center of lower boundary.

The force–displacement curves for both MMs-TO-PSLs and uniform samples during three-point bend tests are presented in Fig. 29. On the whole, it can be seen the experimental results are in good agreement with numerical simulations in Fig. 23. The experimented force–displacement curves can be further divided into elastic, ductile, and fracture stages. The elastic stage is extracted from the near-linear slope of curve. In elastic stage, it can be seen the steep gradient of force–displacement, which indicates nearly no serious local deformation for truss- and plate-based MMs-TO-PSLs structures. It can be seen the elastic strength of plate-based structure is larger than its truss-based counterpart.

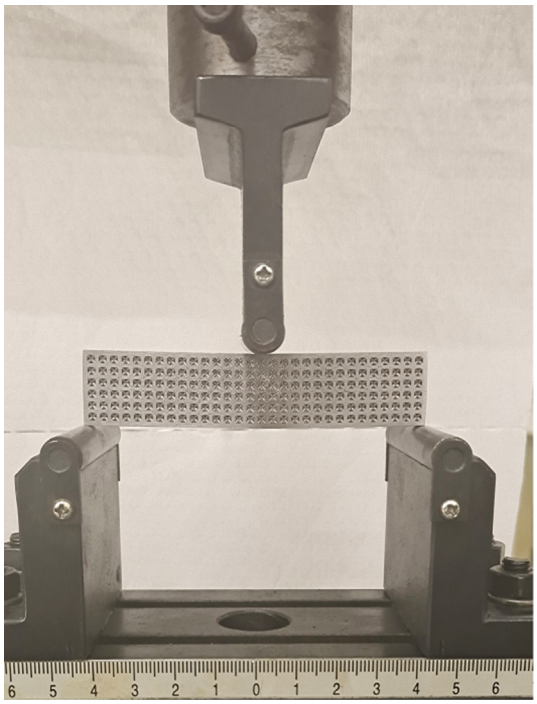


Fig. 27. Three-point bend test equipment and samples.

This is in accordance with our elastoplastic simulations in Section 3.3 and Section 6.2. In plastic deformation stage, the ultimate force occurs before the lower solid region ruptures. The experimented force–displacement curve for MMs mapped uniform sample exhibits a long flat stage, which is induced by severe local deformation around the load applied region. Compared with plate-based MMs-TO-PSLs structure, plastic deformation becomes serious for truss-based MMs-TO-PSLs structure, which leads to

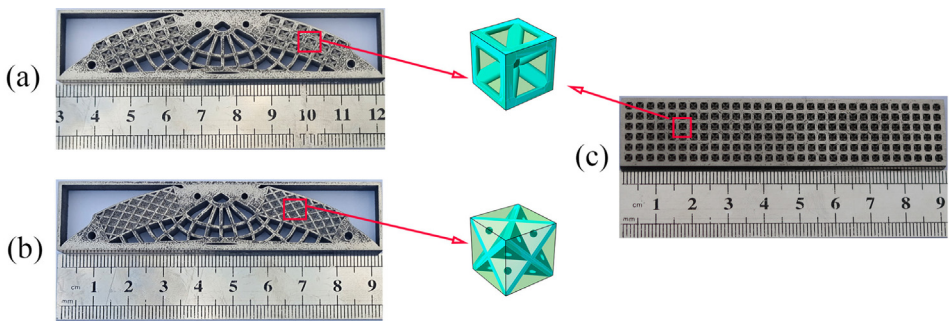


Fig. 26. MMs mapped samples manufactured by SLM system: a) truss-based MMs-TO-PSLs sample, b) plate-based MMs-TO-PSLs sample, c) truss-based MMs mapped uniform sample.

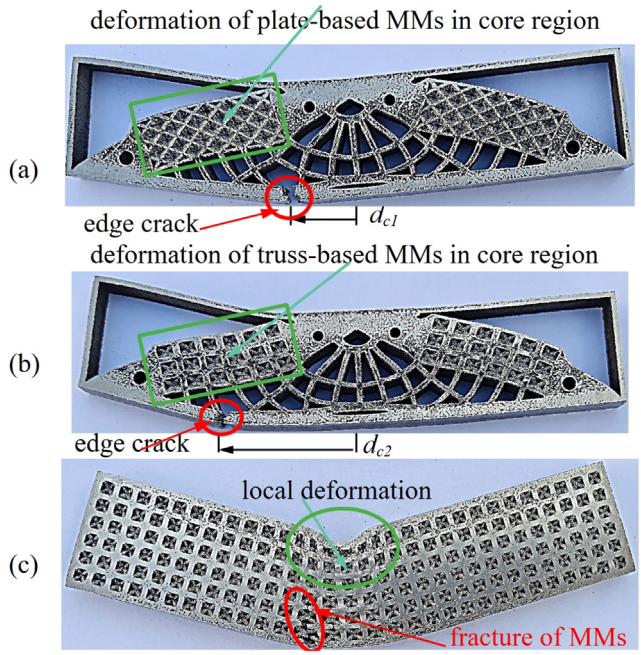


Fig. 28. Fractured MMs mapped samples after three-point bend tests. a) plate-based MMs-TO-PSLs sample, b) truss-based MMs-TO-PSLs sample, c) truss-based MMs mapped uniform sample.

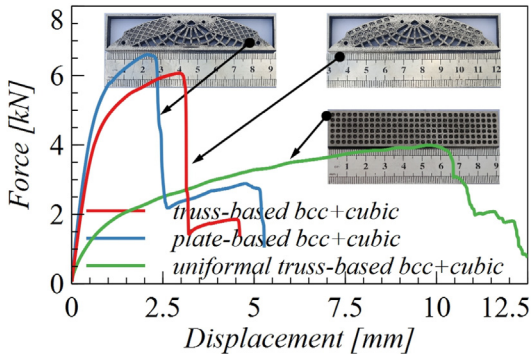


Fig. 29. Displacement-force curves for MMs-TO-PSLs and uniform samples during three-point bend experiment process. The dark blue, red, and green color curves are respectively corresponding to plate-based MMs-TO-PSLs sample, truss-based MMs-TO-PSLs sample, and truss-based MMs infilled uniform sample.

strength degradation. In fracture stage, the drop-off the curve indicates the fracture around the lower boundary. For truss- and plate-based MMs-TO-PSLs samples, the force sharply dropped-off, due to the sudden destroy of the lower solid region. This process is reasonably predicated by ductile failure simulations in Fig. 23. The over drop-off of force results from instantaneous relaxation of stress. Then, new equilibrium reached before next rupture of internal ribbon. The successive raptures of internal mesoscale MMs and ribbons result in fluctuations of force–displacement curve after the first drop of force curve.

8. Conclusion

This work proposes a multiscale lightweight structure design method. Mesoscale mechanical properties of MMs, TO simulations, PSLs, and stress concentration are comprehensively studied. To find out the optimal material distribution, the mesoscale material models are used in TO process and rapid converge is obtained. In mapping process, MMs are filled in the compression-dominated core-region, the [100] directions of MMs are aligned with the normal directions of PSLs, so that near-isotropic mechanical response can be achieved. In addition, the lower bound of machining resolution is considered so that the samples can be successfully fabricated by SLM equipment. It is a fast approach to design lightweight structures as well as to perverse sufficient strength. The MMs-TO-PSLs structure is of programmable mechanical features, due to the mechanical properties can be tuned by the replacing of MMs.

Both plate- and truss-based MMs-TO-PSLs are designed and manufactured by SLM equipment. For a comparison purpose, the conventional MMs mapped uniform sample is also manufactured. The elastoplastic properties and ductile failure evolution of multiscale lightweight structures are investigated by both numerical simulations and mechanical tests. The results show that the strength of truss-based MMs-TO-PSLs sample can be 40% larger than conventional MMs mapped uniform sample in ultimate compression force. Moreover, the plate-based MMs-TO-PSLs structure even can be stronger than its truss-based counterpart.

In mesoscale view, though mechanical anisotropic MMs are applied to MMs-TO-PSLs samples, the [100] orientations of MMs are aligned with the normal directions of PSLs. Under such condition, the shear effect is eased and near isotropic mechanical response can be obtained in core-regions. Due to strong intersection of internal constituting components, the strength of MMs-TO-PSLs sample can be further improved by using plate-based MMs. In agreement with simulations, tensile stress concentrated positions are fractured in experimental tests. Both numerical and experimental results suggest that three stages, including elastic, plastic, and fracture evolution, are experienced during three-point bend process. The MMs-TO-PSLs samples exhibit high ultimate strength. The conventional MMs mapped sample experienced a long flat period, due to strength degradation induced by severe local ductile deformation around load applied region. In fracture evolution stage, the successive fracture of MMs and ribbons results in several jumps of displacement-force curve. The over drop-off of force also can be observed when fracture happens, due to the sudden stress relaxation and dynamic movement of local materials.

There are some limitations that will be considered in future works. The mapped MMs are of weak mechanical-anisotropy. In order to fully ease mechanical-anisotropy, the mechanical-isotropy MMs can be mapped into core-regions. At the same time, though the low density intermediate-region can be infilled by MMs in design process, it is usually impractical to be fabricated by SLM equipment due to the limitation of machining resolution. In present paper, ribbons are infilled in such regions according to PSLs. In future investigations, the characteristic size of elements within MMC will be studied with considering the machinability of SLM equipment.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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